PART A — QUESTIONS

AF8-001
Question:
Solve: $10x - 6 = 64$
A) 6
B) 7
C) 8
D) 9

AF8-002
Question:
If $8n + 3 = 51$, what is n?
A) 5
B) 6
C) 7
D) 8

AF8-003
Question:
Simplify: $5(2x - 3) + 4x$
A) $10x - 15$
B) $14x - 15$
C) $14x - 3$
D) $6x - 15$

AF8-004
Question:
A company charges $22 per visit plus $9 for each supply used. Which expression represents the total charge for s supplies?
A) $22s + 9$
B) $9s + 22$
C) $31s$
D) $22 - 9s$

AF8-005
Question:
If $f(x) = 4x - 11$, what is $f(8)$?
A) 19
B) 21
C) 32
D) 43

AF8-006
Question:
Which ordered pair satisfies the equation $y = 3x - 7$?
A) (3, 1)
B) (4, 5)
C) (5, 8)
D) (6, 12)

AF8-007
Question:
Solve: $9(x - 3) = 45$
A) 6
B) 7
C) 8
D) 9

AF8-008
Question:
A pattern begins: 11, 18, 25, 32, ... What is the next number?
A) 37
B) 38
C) 39
D) 40

AF8-009
Question:
Solve: $x/14 = 3$
A) 17
B) 28
C) 42
D) 56

AF8-010
Question:
A line has a slope of -6 and crosses the y-axis at 5. Which equation represents the line?
A) $y = 5x - 6$
B) $y = -6x + 5$
C) $y = 6x + 5$
D) $y = -5x + 6$

AF8-011
Question:
If 3 bundles contain 57 pieces, how many pieces are in 8 bundles at the same rate?
A) 128
B) 144
C) 152
D) 171

AF8-012
Question:
Solve: $42 - 7x = 14$
A) 3
B) 4
C) 5
D) 6

AF8-013
Question:
Which expression is equivalent to $24d - 9d - 8$?

A) 15d - 8
B) 33d - 8
C) 15d + 8
D) 24d - 17

AF8-014
Question:
A shelf starts with 95 items. Workers remove 6 items each hour. Which equation gives the number of items I left after h hours?
A) I = 95h - 6
B) I = 6h + 95
C) I = 95 - 6h
D) I = 6h - 95

AF8-015
Question:
If y = -7x + 30, what is y when x = 4?
A) 2
B) 7
C) 26
D) 58

AF8-016
Question:
Solve: 13x + 9 = 7x + 45
A) 5
B) 6
C) 7
D) 8

AF8-017
Question:
Which value of x makes 6x - 2 less than 34?
A) 5
B) 6
C) 7
D) 8

AF8-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 7, 15, 23, 31

What is the rule?
A) y = 7x + 8
B) y = 8x + 7
C) y = 15x - 8
D) y = x + 7

AF8-019
Question:
Solve: $5(x + 6) = 2x + 48$
A) 4
B) 5
C) 6
D) 7

AF8-020
Question:
If $m = -8$ and $n = 3$, what is $m + 4n$?
A) -20
B) -5
C) 4
D) 20

AF8-021
Question:
Which point is located in Quadrant I?
A) (-3, 4)
B) (3, -4)
C) (-3, -4)
D) (3, 4)

AF8-022
Question:
A worker sorts 72 items in 8 minutes. At the same rate, how many items can the worker sort in 6 minutes?
A) 48
B) 54
C) 60
D) 66

AF8-023
Question:
Solve for r: $H = 8r + 24$
A) $r = H - 24$
B) $r = (H - 24)/8$
C) $r = 8(H - 24)$
D) $r = H/8 + 24$

AF8-024
Question:
What is the slope of a line that goes through (1, 6) and (5, 26)?
A) 4
B) 5
C) 6
D) 20

AF8-025
Question:

Simplify: $17x + 3 - 9x - 15$
A) $8x - 12$
B) $26x - 12$
C) $8x + 18$
D) $26x + 18$

AF8-026
Question:
If $g(x) = x^2 + 3x$, what is $g(6)$?
A) 36
B) 42
C) 54
D) 72

AF8-027
Question:
A sequence follows the rule “add 6, then multiply by 2.” If the first term is 7, what is the second term?
A) 13
B) 19
C) 26
D) 28

AF8-028
Question:
Solve: $8(x - 2) + 4 = 52$
A) 5
B) 6
C) 7
D) 8

AF8-029
Question:
A line crosses the y-axis at -9 and rises 3 units for every 1 unit to the right. Which equation represents the line?
A) $y = -9x + 3$
B) $y = 3x - 9$
C) $y = -3x - 9$
D) $y = 9x + 3$

AF8-030
Question:
If $10/12 = x/36$, what is x?
A) 24
B) 27
C) 30
D) 32

AF8-031
Question:
Solve: $-10x + 15 = -65$

A) -8
B) -5
C) 5
D) 8

AF8-032
Question:
A bin starts with 600 clips. Each package uses 25 clips. Which expression gives the number of clips left after p packages?
A) 600 + 25p
B) 600 - 25p
C) 25p - 600
D) 600 ÷ 25p

AF8-033
Question:
Which equation has x = 13 as its solution?
A) 2x + 9 = 33
B) 3x - 5 = 34
C) x/13 + 4 = 6
D) 4x - 7 = 47

PART B — ANSWER KEY + EXPLANATIONS

AF8-001 — Correct: B — Explanation:
Solve 10x - 6 = 64. Add 6 to both sides: 10x = 70. Divide by 10: x = 7. The most common mistake is subtracting 6 instead of adding 6.

AF8-002 — Correct: B — Explanation:
Start with 8n + 3 = 51. Subtract 3 from both sides: 8n = 48. Divide by 8: n = 6. A common mistake is stopping at 48 and forgetting to divide by 8.

AF8-003 — Correct: B — Explanation:
Distribute first: 5(2x - 3) = 10x - 15. Then add 4x: 10x + 4x - 15 = 14x - 15. A common mistake is forgetting to multiply -3 by 5.

AF8-004 — Correct: B — Explanation:
The company charges $9 for each supply, so that part is 9s. The $22 visit charge is added one time. The total charge is 9s + 22. A common mistake is treating the visit charge as if it repeats for every supply.

AF8-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 4(8) - 11 = 32 - 11 = 21. A common mistake is adding 11 instead of subtracting it.

AF8-006 — Correct: C — Explanation:
Test the choices in y = 3x - 7. For (5, 8), 3(5) - 7 = 15 - 7 = 8, which matches the y-value. A common mistake is choosing a point that looks close without checking both sides.

AF8-007 — Correct: C — Explanation:
Solve $9(x - 3) = 45$. Divide both sides by 9: $x - 3 = 5$. Add 3: $x = 8$. A common mistake is adding 3 before undoing the multiplication by 9.

AF8-008 — Correct: C — Explanation:
The pattern increases by 7 each time: 11, 18, 25, 32. Add 7 to 32: 39. A common mistake is assuming the next number is 38 because the increase seems close to 6.

AF8-009 — Correct: C — Explanation:
$x/14 = 3$ means x divided by 14 equals 3. Multiply both sides by 14: $x = 42$. A common mistake is adding 14 and 3 instead of multiplying.

AF8-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -6 and the y-intercept is 5, so the equation is $y = -6x + 5$. A common mistake is switching the slope and y-intercept.

AF8-011 — Correct: C — Explanation:
Find the unit rate: 57 pieces ÷ 3 bundles = 19 pieces per bundle. For 8 bundles: $19 \times 8 = 152$. A common mistake is adding 5 bundles and 5 pieces instead of using the rate.

AF8-012 — Correct: B — Explanation:
Solve $42 - 7x = 14$. Subtract 42 from both sides: $-7x = -28$. Divide by -7: $x = 4$. A common mistake is losing the negative sign after subtracting 42.

AF8-013 — Correct: A — Explanation:
Combine like terms: $24d - 9d = 15d$. The constant -8 stays the same. The expression becomes $15d - 8$. A common mistake is combining the constant with the variable terms.

AF8-014 — Correct: C — Explanation:
The shelf starts with 95 items. Workers remove 6 items each hour, so after h hours, 6h items are removed. The number left is $I = 95 - 6h$. A common mistake is adding 6h even though items are being removed.

AF8-015 — Correct: A — Explanation:
Substitute $x = 4$ into $y = -7x + 30$: $y = -7(4) + 30 = -28 + 30 = 2$. A common mistake is treating $-7(4)$ as positive 28.

AF8-016 — Correct: B — Explanation:
Solve $13x + 9 = 7x + 45$. Subtract 7x: $6x + 9 = 45$. Subtract 9: $6x = 36$. Divide by 6: $x = 6$. A common mistake is subtracting 9 from only one side.

AF8-017 — Correct: A — Explanation:
Solve $6x - 2 < 34$. Add 2: $6x < 36$. Divide by 6: $x < 6$. The only listed value less than 6 is 5. A common mistake is choosing 6, but $6(6) - 2 = 34$, not less than 34.

AF8-018 — Correct: B — Explanation:
The y-values increase by 8 each time x increases by 1, so the slope is 8. When $x = 0$, $y = 7$, so the y-intercept is 7. The rule is $y = 8x + 7$. A common mistake is using 7 as the slope because it is the first y-value.

AF8-019 — Correct: C — Explanation:
Distribute: $5x + 30 = 2x + 48$. Subtract 2x: $3x + 30 = 48$. Subtract 30: $3x = 18$. Divide by 3: $x = 6$. A common mistake is forgetting to multiply 6 by 5.

AF8-020 — Correct: C — Explanation:
Substitute $m = -8$ and $n = 3$: $m + 4n = -8 + 4(3) = -8 + 12 = 4$. A common mistake is treating -8 as positive 8.

AF8-021 — Correct: D — Explanation:
Quadrant I has a positive x-coordinate and a positive y-coordinate. The point (3, 4) fits this description. A common mistake is choosing a point with only one positive coordinate.

AF8-022 — Correct: B — Explanation:
72 items in 8 minutes means $72 \div 8 = 9$ items per minute. In 6 minutes: $9 \times 6 = 54$. A common mistake is multiplying 72 by 6 without finding the rate first.

AF8-023 — Correct: B — Explanation:
Start with $H = 8r + 24$. Subtract 24 from both sides: $H - 24 = 8r$. Divide by 8: $r = (H - 24)/8$. A common mistake is dividing H by 8 first and then subtracting 24.

AF8-024 — Correct: B — Explanation:
Slope equals change in y divided by change in x. From (1, 6) to (5, 26), y increases by 20 and x increases by 4. The slope is $20 \div 4 = 5$. A common mistake is using 20 as the slope without dividing by the change in x.

AF8-025 — Correct: A — Explanation:
Combine like terms: $17x - 9x = 8x$, and $3 - 15 = -12$. The simplified expression is $8x - 12$. A common mistake is writing $8x + 12$ because the negative sign before 15 is missed.

AF8-026 — Correct: C — Explanation:
Substitute 6 for x: $g(6) = 6^2 + 3(6) = 36 + 18 = 54$. A common mistake is calculating 6^2 as 12 instead of 36.

AF8-027 — Correct: C — Explanation:
Start with 7. Add 6: 13. Then multiply by 2: 26. A common mistake is multiplying first and then adding 6.

AF8-028 — Correct: D — Explanation:
Solve $8(x - 2) + 4 = 52$. Distribute: $8x - 16 + 4 = 52$. Combine: $8x - 12 = 52$. Add 12: $8x = 64$. Divide by 8: $x = 8$. A common mistake is forgetting to combine -16 and +4.

AF8-029 — Correct: B — Explanation:
The line rises 3 units for every 1 unit to the right, so the slope is 3. It crosses the y-axis at -9, so the equation is $y = 3x - 9$. A common mistake is switching the slope and intercept.

AF8-030 — Correct: C — Explanation:
Use the proportion $10/12 = x/36$. Since $12 \times 3 = 36$, multiply 10 by 3: $x = 30$. A common mistake is adding 24 to the numerator because 12 becomes 36.

AF8-031 — Correct: D — Explanation:
Solve -10x + 15 = -65. Subtract 15 from both sides: -10x = -80. Divide by -10: x = 8. A common mistake is giving -8 because both sides contain negative numbers.

AF8-032 — Correct: B — Explanation:
The bin starts with 600 clips. Each package uses 25 clips, so p packages use 25p clips. The number left is 600 - 25p. A common mistake is adding 25p even though clips are being used.

AF8-033 — Correct: B — Explanation:
Test x = 13. In choice B, 3(13) - 5 = 39 - 5 = 34, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.